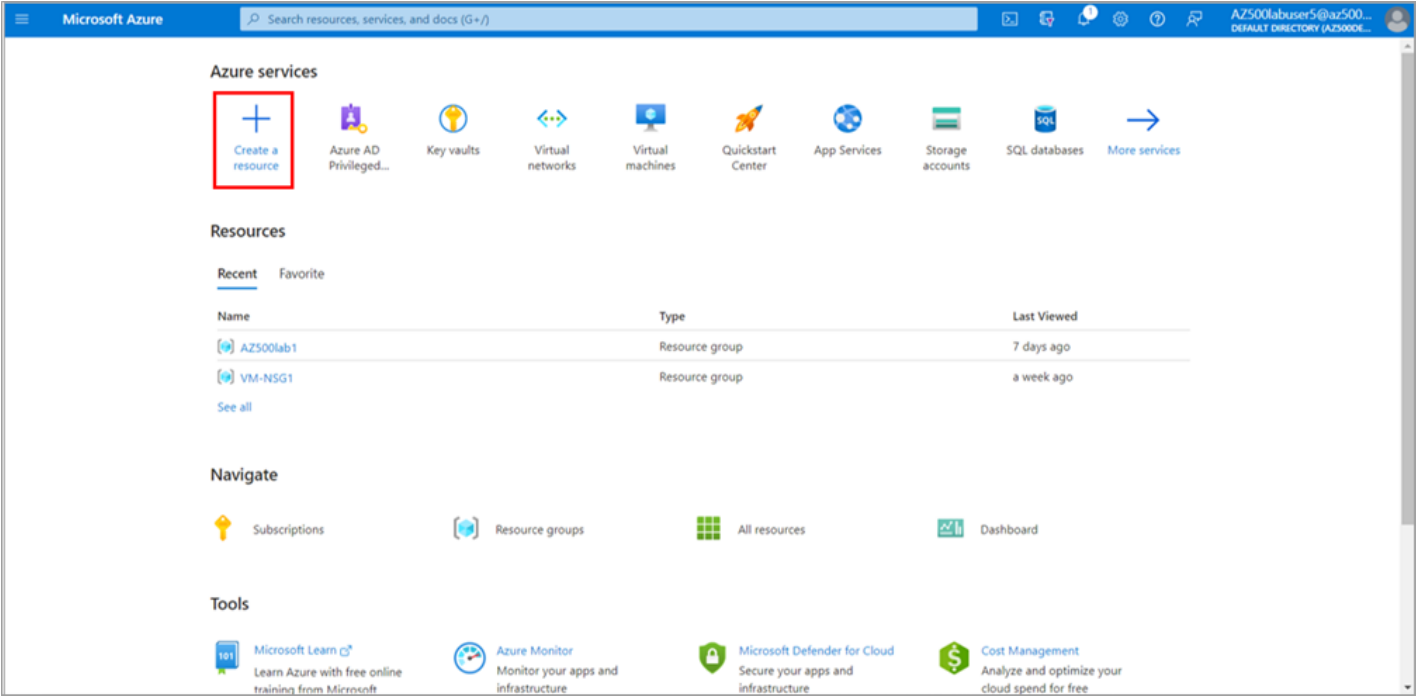


Key Vault project solution

Instructions: List the steps you will perform to complete each task in the system. Ensure to provide screenshots for each step. As a sample, we have filled in the information for the first two steps in *Task 1, Activity 1*. We have used red highlights to show how relevant the screenshot is for the step. However, you do not need to use any highlights in your work. You can add more rows as required.

Task 1: Store a secret in a Key Vault and use it during virtual machine creation.

In this task, you store a secret, which is a password for your Windows virtual machines (VMs).

Activity #	Steps
Activity 1: Create a Key Vault.	1. Sign in to the Azure portal with your login credentials. Select Create a resource.  2. In the Search box, enter Key Vault. From the results list, select Key Vault.

Microsoft Azure

Search resources, services, and docs (G+7)

AZ500labuser5@az500...
DEFAULT DIRECTORY (AZ500DE...)

[Home](#) > [Create a resource](#) >

Marketplace

Get Started

Service Providers

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

Recently created

Private products

Categories

Security (30)

DevOps (18)

Identity (16)

Analytics (10)

Compute (10)

Key Vault

Pricing : All

Operating System : All

Publisher Type : All

Product Type : All

Publisher name : All

Showing 1 to 20 of 80 results for 'Key Vault'. [Clear search](#)

Tile view



Key Vault

Microsoft

Azure Service

Safeguard cryptographic keys and other secrets used by cloud apps and services.

Create



KoçSistem Azure Key Vault Management

KoçSistem Bilgi ve İletişim Hizm...

Managed Services

Increase security and control over keys and passwords with the support of Azure Key Vault Services!

Create



Azure Key Vault solution for Sentinel

Microsoft Sentinel, Microsoft Co...

Azure Application

Azure Key Vault solution for Sentinel

Price varies

Create



Azure Key Vault Managed HSM

Microsoft

Azure Service

Safeguard cryptographic keys used by cloud apps and services.

Create



EJBCA SaaS - With Key Vault Backed CA Keys

Keyfactor, Inc.

SaaS

EJBCA SaaS. PKI delivered as a service with Azure Key Vault key storage

Starts at
₺388,617.275/month

Subscribe



HashiCorp Vault Enterprise

HashiCorp

SaaS



Hashicorp Vault on CentOS

AskforCloud LLC

Virtual Machine



EJBCA SaaS - With Azure Key Vault Managed HSM

Keyfactor, Inc.

SaaS



Vault on Debian 10

AskforCloud LLC

Virtual Machine



Vault on Debian 11

AskforCloud LLC

Virtual Machine

Is Marketplace helpful?

3. Keyvault has been created.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a user profile for 'nicepants77@gmail.com'. The left sidebar lists various Azure services, with 'Key vault' selected. The main content area shows the 'secret-kv-new' Key Vault overview. The 'Overview' tab is active, displaying essential information: Resource group (Ungraded-keyvault), Location (East US), Subscription (Azure subscription 1), and Subscription ID (ae057139-3ee2-4662-a3fb-11c8432a3437). The 'Essentials' section lists properties like Vault URI, Sku (Standard), Directory ID, Directory Name, Soft-delete (Enabled), and Purge protection (Disabled). Below this, a 'Get started' section provides guidance on managing keys and secrets, recommending a vault per application per environment. At the bottom, three action cards are visible: 'Control access to key vault', 'Enable logging and set up alerts', and 'Turn on recovery options'.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Ungraded-keyvault

secret-kv-new Key vault

Show me overall vault saturation percentage metrics of this Key Vault. Show me total service API hits metrics of this Key Vault. +1

Search Delete Move Refresh Open in mobile

Overview

Activity log Access control (IAM) Tags Diagnose and solve problems Access policies Resource visualizer Events Objects Settings Monitoring Automation Help

Essentials

Resource group (move) Ungraded-keyvault Location East US Subscription (move) Azure subscription 1 Subscription ID ae057139-3ee2-4662-a3fb-11c8432a3437

Tags (edit) Add tags

Get started Properties Monitoring Tools + SDKs Tutorials

Manage keys and secrets used by apps and services

Our recommendation is to use a vault per application per environment (Development, Pre-Production and Production). This helps you to not share secrets across environments and also reduces the threat in case of a breach.

Control access to key vault Enable logging and set up alerts Turn on recovery options

Activity 2:

Save a secret in Azure Key Vault.

1. Saving a secret in the key vault

The screenshot shows the Azure portal interface for creating a new secret in a Key Vault. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a Copilot icon. The user's profile 'nicepants77@gmail.com' is visible in the top right. The left sidebar contains a navigation menu with options like 'Create a resource', 'Home', 'Dashboard', 'All services', and a 'FAVORITES' section listing various Azure services. The main content area is titled 'Create a secret' and shows the breadcrumb path 'Home > Ungraded-keyvault > secret-kv-new | Secrets'. The form includes fields for 'Upload options' (set to 'Manual'), 'Name' (set to 'Password'), and 'Secret value' (masked with dots). There are checkboxes for 'Content type (optional)', 'Set activation date', and 'Set expiration date'. The 'Enabled' toggle is set to 'Yes', and the 'Tags' section shows '0 tags'. At the bottom, there are 'Create' and 'Cancel' buttons.

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot nicepants77@gmail.com DEFAULT DIRECTORY (NICEPANT...

Home > Ungraded-keyvault > secret-kv-new | Secrets

Create a secret

Upload options: Manual

Name: Password

Secret value:

Content type (optional):

Set activation date: ☐

Set expiration date: ☐

Enabled: Yes No

Tags: 0 tags

Create Cancel

2. Made myself a Key Vault Administrator to manage secrets

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a Copilot button. The user's profile 'nicepants77@gmail.com' is in the top right. The left sidebar contains a navigation menu with options like 'Create a resource', 'Home', 'Dashboard', 'All services', and a 'FAVORITES' section listing various Azure services. The main content area displays the 'Add role assignment' dialog for the 'secret-kv-new' resource under 'Access control (IAM)'. The dialog has tabs for 'Role', 'Members', 'Conditions', and 'Review + assign'. The 'Review + assign' tab is active, showing the role 'Key Vault Administrator' and its scope. A table lists the assigned member 'Charlie Chaplin(Guest)' with their object ID and user type. At the bottom, there are buttons for 'Review + assign', 'Previous', and 'Next', along with a 'Feedback' link.

Home > secret-kv-new | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

Role Key Vault Administrator

Scope /subscriptions/ae057139-3ee2-4662-a3fb-11c8432a3437/resourceGroups/Ungraded-keyvault/providers/Microsoft.KeyVault/vaults/secret-kv-new

Members	Name	Object ID	Type
	Charlie Chaplin(Guest)	2f043ba1-7fc9-4aa4-a2f6-362f02b6a886	User

Description No description

[Review + assign](#) [Previous](#) [Next](#) [Feedback](#)

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/)

Copilot

2

nicepants77@gmail.com
DEFAULT DIRECTORY (NICEPANT...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > secret-kv-new

secret-kv-new | Secrets ☆

Key vault

Search

Generate/Import Refresh Restore Backup Manage deleted secrets View sample code

The secret 'Password' has been successfully created.

Name	Type	Status	Expiration date
Password		✓ Enabled	

OverviewActivity logAccess control (IAM)TagsDiagnose and solve problemsAccess policiesResource visualizerEventsObjectsKeysSecretsCertificatesSettingsMonitoringAutomationHelp

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Activity 3:
Create a DevTest Lab.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

2

nicepants77@gmail.com

DEFAULT DIRECTORY (NICEPANT...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > DevTest Labs

Dev-ungradedlab

DevTest Lab

Search

Overview

Getting started

Internal support

Resource visualizer

My Lab

Claimable virtual machines

All resources

My virtual machines

My environments

My data disks

My secrets

Settings

Generate a query for this resource

Are there any outages impacting this resource

Query dependencies for this resource

Upcoming Retirements on Sept 30, 2025:

Basic Load Balancers and Public IPs: Upgrade existing Basic Load Balancers and Public IPs to Standard SKU. To avoid disruptions post-upgrade, do not rename the load balancer during the upgrade. For more information, see the [official announcement](#) or read our [upgrade by name guidance](#).

Default Outbound Access for Private IP VMs: Find out how you can [configure your outbound access here](#) for your VMs to ensure continued access.

Essentials

Resource group [\(move\)](#)

Ungraded-keyvault

Status

Ready

Location

East US

Subscription [\(move\)](#)

Azure subscription 1

Subscription ID

ae057139-3ee2-4662-a3fb-11c8432a3437

My virtual machines

Virtual machines you have added and claimed appear in this list. Choose "claim any" to immediately claim a random virtual machine, or see [Claimable virtual machines](#) to choose among all existing machines available in this DevTest Lab.

Name ↑↓	Status ↑↓	Auto-start ↑↓	Auto-shutdown ↑↓	Base ↑↓
Nothing to display				

Add or remove favorites by pressing Ctrl+Shift+F

Activity 4:
Create a virtual machine in DevTest Lab.

1. Setting up a VM in DevTest Lab

Basic Settings

Advanced Settings

Tags

User Settings

Virtual machine name *

TheGreatVM

✓

User name *

HxVM

✓

Use a saved secret ⓘ

☐

Password *

.....

✓

Save as default password

☐

Disk and Size

Image Base

Windows 11 Enterprise multi-session + Microsoft 365 Apps, version 22H2 - x64 Gen2

Virtual machine size * ⓘ

Size

Standard_D4s_v5

[Change Size](#)

Allow hibernation ⓘ

☐

OS disk type ⓘ

Standard SSD

▼

Security features

Security type ⓘ

Standard

Trusted launch

Artifacts

Artifacts

Artifact

0 artifact(s) configured

[Add or Remove Artifacts](#)

2. Added chrome as an artifact in the VM as stated in the *Project Requirements*.

 Applying artifacts on small sized (1 core) Windows VMs may take a longer time or fail in certain cases

Select the artifacts to be installed on the VM. Artifacts will be installed in the order chosen, from top to bottom. [Learn more.](#)

1 artifact(s) configured

 Search to filter available artifacts

Available artifacts

Name

Loading...

Selected artifacts

Name

 Chrome



Atom	>
 AWS Command Line Interface	>
 Azure Pipelines Agent	>
 Azure Pipelines Agent(s)	>
 Azure Pipelines Deployment Group Agent	>
 Azure PowerShell	>
 Beyond Compare	>
 Chef Client	>
 Chrome	>
 Clone a git repository	>
 Configuration Manager Current Branch	>
 Configuration Manager Technical Preview	>
 Configure WinRM	>
 Create Domain	>

Activity 5:
Complete the virtual machine deployment.

1. VM deployment **failed** due to an error on Azure’s side.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

nicepants77@gmail.com
DEFAULT DIRECTORY (NICEPANT...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Ungraded-keyvault

vm-lab | Overview

virtual machine

Search

Connect

Start

Restart

Stop

Hibernate

Artifacts

Claim machine

Unclaim

Delete

Refresh

General

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Operations

Create custom image

Create formula (reusable base)

Artifacts

Auto-shutdown

Auto-start

Redeploy

Expiration date

Settings

Disks

Security

Add or remove favorites by pressing Ctrl+Shift+F

Essentials

Status

Failed

Resource group

dev-ungradedlab984571964000

Virtual network/subnet

DtlDev-ungradedlab/DtlDev-ungradedlabSubnet

IP address or FQDN

10.0.0.5

NAT protocol / Port to connect

Private

Size

Standard_D4s_v5

Operating system

Windows

Base

Windows 11 Enterprise, version 25H2 (x64) (Microsoft Dev Box Compati...

Expiration date

No expiration

Security type

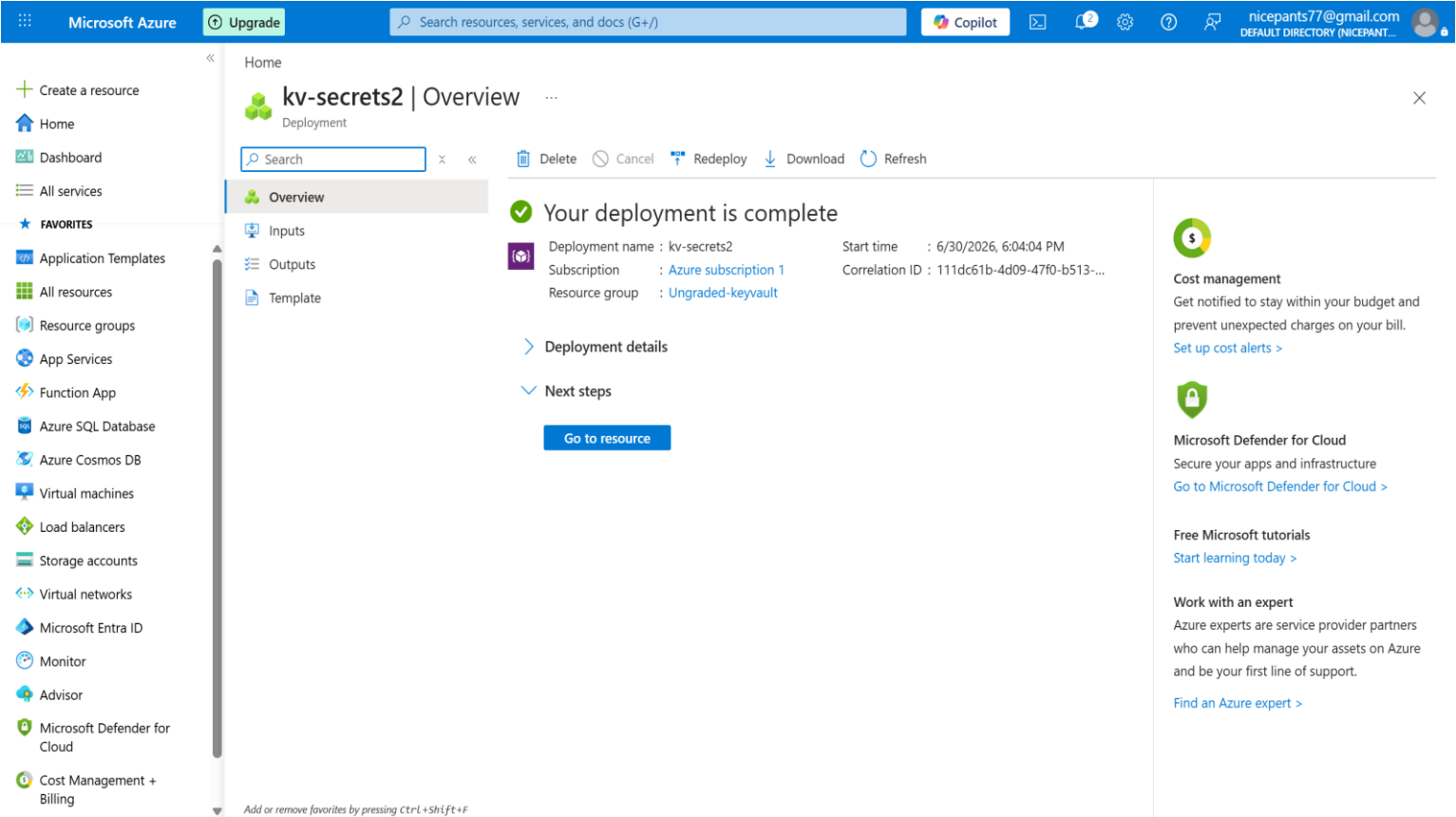
Standard

Schedules

Name	Status	Time	Time zone	Day of week
Auto-start	Opted-out			
Auto-shutdown	Opted-out			

Task 2: Create a key in a Key Vault.

In this task, you create a key named *Key1* in a Key Vault named *Key-vaults-secret*.

Activity #	Steps
Activity: Create a key.	1. Creating the key vault named <i>kv-secret2</i> 

2. I made myself the Key Vault Administrator to manage keys

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an Upgrade button, a search bar, and the Copilot icon. The user's profile is visible in the top right corner.

The left sidebar contains a navigation menu with options like Create a resource, Home, Dashboard, All services, and a Favorites section. The main content area is titled "kv-secrets2 | Access control (IAM)" and shows the "Access control (IAM)" tab selected. A warning message at the top states: "Action required: 1 user has elevated access in your tenant. You should take immediate action and remove all role assignments with elevated access. In addition, remove any role assignments at root scope that are no longer needed. View role assignments".

The "Role assignments" tab is active, showing a table of role assignments for the subscription. The table has columns for Name, Type, Role, Scope, and Condition. The assignments are grouped by role: Owner (2), Key Vault Administrator (1), and User Access Administrator (1).

Name	Type	Role	Scope	Condition
Owner (2)				
Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Owner	Subscription (Inherited)	None
Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Owner	Subscription (Inherited)	None
Key Vault Administrator (1)				
Charlie Chaplin 2f043ba1-7fc9-4aa...	User	Key Vault Administrator	This resource	None
User Access Administrator (1)				
Charlie Chaplin				

3. Creating the key with rotation policy for auto-rotation of the key's value.

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot nicepants77@gmail.com DEFAULT DIRECTORY (NICEPANT...

Home > kv-secrets2 | Keys

Create a key

Options Generate

Name * ⓘ KeyToHeaven

Key type ⓘ ☒ RSA ☐ EC

RSA key size ☒ 2048 ☐ 3072 ☐ 4096

Set activation date ⓘ ☐

Set expiration date ⓘ ☐

Enabled ☒ Yes ☐ No

Tags 0 tags

Set key rotation policy Not configured

Confidential Key Options

Exportable ⓘ ☐

Immutable ⓘ ☐

Create Cancel

Rotation policy

KeyToHeaven

Expiry time 30 days ✓

Rotation

Enable auto rotation ☒ Enabled ☐ Disabled

Rotation option ⓘ Automatically renew at a given time after c... ✓

Rotation time 10 days ✓

Notification

Notification option ⓘ Notify at a given time before expiry

Notification time 30 days

OK

4. The key is created **successfully**.

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot nicepants77@gmail.com DEFAULT DIRECTORY (NICEPANT...

Home > kv-secrets2

kv-secrets2 | Keys ☆ ...

Key vault

Search + Generate/Import Refresh Restore Backup Manage deleted keys

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Access policies Resource visualizer Events Objects

Keys Secrets Certificates Settings Monitoring Automation Help

The key 'KeyToHeaven' has been successfully created.

Name	Status	Expiration date
KeyToHeaven	✓ Enabled	

Updating the key 'KeyToHeaven'.
The key 'KeyToHeaven' has been successfully updated.

Creating the key 'KeyToHeaven'.
The key 'KeyToHeaven' has been successfully created.

Create a resource Home Dashboard All services FAVORITES Application Templates All resources Resource groups App Services Function App Azure SQL Database Azure Cosmos DB Virtual machines Load balancers Storage accounts Virtual networks Microsoft Entra ID Monitor Advisor Microsoft Defender for Cloud Cost Management + Billing

Add or remove favorites by pressing Ctrl+Shift+F